

OpenVLA-OFT · arXiv 2502.19645 · Kim, Finn, Liang (Stanford)

Visionary

11+3

Harness vs BEHAVIOR deploy
 $\pi_{0.5}$ /OpenPI primary · no Discord/Drive

Oct 26 paper club · VLA Architectures

SPINE: AR discrete bins → parallel continuous L1 chunks

OFT $\neq$ FAST $\neq$ FM · FT recipe / speed+success patch · not $\pi_{0.5}$ on BEHAVIOR

HARNESS vs BEHAVIOR DEPLOY

Visionary · 2026 — no Discord/Drive

HARNESS (keep OFT)

- Open AR-family control / ablation
- Parallel continuous chunk FT moral
- L1 specialist adaptation (fast)
- FiLM lesson for multi-cam lang
- Measure throughput AND latency
- Stock AR bins → museum piece

DEPLOY ($\pi_{0.5}$ / OpenPI)

- 2026 BEHAVIOR primary substrate
- FM(+RTC/KI) continuous chunks
- Stages > monolithic FT
- Multimodal / recovery heads
- Household open-world coverage
- OFT weights $\neq$ entrant

Adopt OFT morals, not OFT weights

Primary head likely FM ($\pi_{0.5}$ / N1.7), not 7B L1 MLP

Skip Discord/Drive logistics · LIBERO 97.1% $\neq$ BEHAVIOR q-score

TAXONOMY · $\text{OFT} \neq \text{FAST} \neq \text{FM}$

Three parallel improvement paths — do not collapse names

OpenVLA-OFT

FT recipe

Parallel continuous
L1 chunks
K=8 / 25

Decode: 1-pass
Loss: L1
Still 7B OpenVLA

FAST

Tokenizer

DCT→BPE discrete
chunk tokens
still AR

Decode: serial AR
Loss: CE
~750 ms/chunk class

π_0 / $\pi_{0.5}$

FM deploy

Continuous FM
chunks / NFE
OpenPI stack

Decode: denoise
Loss: FM / ϵ
 π_0 ALOHA ~291 Hz

Vanilla OpenVLA = AR discrete bins @ ~4.2 Hz → OFT rewrites output signature

OFT does NOT replace $\pi_{0.5}$ for 2026 BEHAVIOR · $\text{OFT} \neq \text{FAST}$ tokenizer swap

FORK GENEALOGY · OFT || FAST || FM

Archaeologist — parallel improvement paths; do not collapse



STAKEHOLDER ONE-LINER

OFT = speed+success FT patch

making OpenVLA viable at 25–50 Hz

(26–43× throughput, LIBERO 97.1%) — modernize OpenVLA ops,

don't substitute for $\pi_{0.5}$ on BEHAVIOR.

OFT = FT recipe

not a new FM

Not FAST

still-AR tokenizer path

Not FM / $\pi_{0.5}$

generative continuous expert

GitHub title: [ECE 605] - OpenVLA-OFT Making Stakeholder

arXiv 2502.19645 · openvla-oft.github.io

DECISION TREE · OFT vs $\pi_{0.5}$ vs FAST

Stakeholder HIGH priority — say this aloud

Need high-Hz control on OpenVLA stack?

NO

Stock OpenVLA OK for
<10 Hz language demos

YES → adopt OFT

parallel + chunk + Cont-L1

Choose OFT when

- Prismatic / HF / LoRA ops
- Simple L1 FT preferred
- LIBERO-style + moderate demos
- 7B AR-lineage ablations
- Wrist+proprio match robot I/O

Choose $\pi_{0.5}$ when

- OpenPI standardized
- Long-horizon open-world
- 2026 BEHAVIOR primary
- Need multimodal / recovery
- Max wall-clock (291 Hz class)

Choose FAST when

- Tokenizer-only AR fix
- Keep discrete CE head
- Train-speed / compression
- Not high-Hz bimanual alone
- Different lever than OFT

Do NOT pitch OFT as “beats $\pi_{0.5}$ for BEHAVIOR” — best OpenVLA-family FT recipe only

RISKS · staying on OpenVLA family

Stakeholder honesty list

1 Fine-tuning 2024 OXE AR $\neq$ 2025 FM generalist

2 L1 unimodality — weak on true multimodal demos

3 FiLM tax — +456M params; eng complexity

4 ALOHA chunk latency 0.321 s vs π_0 0.086 s

5 “97% LIBERO” $\neq$ BEHAVIOR-ready

6 Put-in-pot $\sim$ 51% — not product-reliable

7 Dual-stack cost if also running $\pi_{0.5}$

8 FAST $\neq$ OFT confusion (different levers)

SIGNATURE SHIFT DIAGRAM

AR discrete single-step → parallel continuous L1 chunks

VANILLA OpenVLA

img + lang → VLM (causal)

AR next-token × D(=7)

$7 \times \{0..255\}$ bins → dequantize

single EE delta · K=1

~4.2 Hz LIBERO · ~1.8 Hz ALOHA

OFT

OpenVLA-OFT

img[+wrist][+proprio] + lang

bidirectional action queries

one forward → $a \in \mathbb{R}^{K \times D}$

continuous L1 MLP · K=8/25

~109.7 Hz · ALOHA ~77.9 Hz

FAST (sibling)

compress chunk → fewer
AR tokens; still sequential
~750 ms chunk class

π_0 / DP / ACT

continuous chunk experts
FM / diffusion / CVAE+L1
smaller bodies → higher Hz

OFT+

OFT + FiLM on both
vision towers
ALOHA multi-view lang

Fig.1 · OFT+ ALOHA system · parallel chunk + FiLM

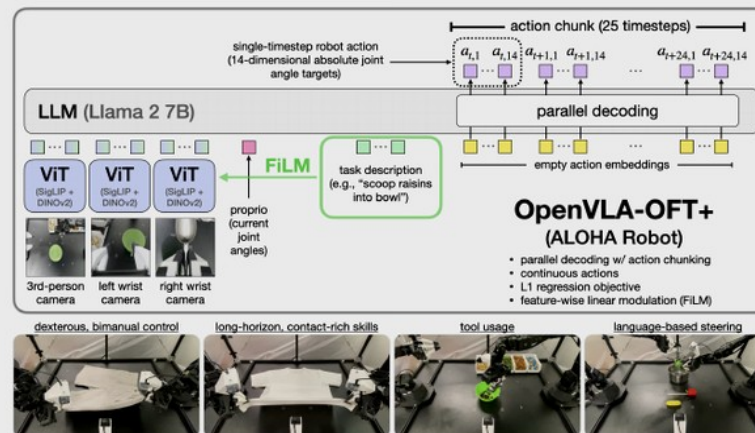


Fig. 1: OpenVLA-OFT+ on the bimanual ALOHA robot. Our Optimized Fine-Tuning (OFT) recipe enhances fine-tuned OpenVLA policies through improved inference efficiency, model quality, and input-output flexibility. The resulting OpenVLA-OFT+ policies execute diverse dexterous manipulation tasks on a real-world bimanual robot at high control frequencies (25 Hz). The “+” suffix indicates the integration of feature-wise linear modulation (FiLM) [37], which strengthens language grounding in tasks where accurate language understanding is critical for success.

denoting the augmented recipe as **OFT+**. OpenVLA-OFT+ successfully executes dexterous bimanual tasks like folding clothes and manipulating target food items based on the user’s prompt (see Figure 1), outperforming both fine-tuned VLAs (π_0 and RDT-1B) and prominent imitation learning policies trained from scratch (Diffusion Policy and ACT) by up to 15% (absolute) in average success rate. With 25-timestep action chunks, OpenVLA-OFT+ achieves $43\times$ faster throughput than base OpenVLA, demonstrating that our new fine-tuning recipe enables real-time robot control with strong task performance and language following ability.

II. RELATED WORK

Prior works have leveraged language and vision foundation models to enhance robotic capabilities, using them as pretrained visual representations that accelerate robotic policy learning [30, 33, 31, 20, 32], for object localization in robotics tasks [9, 47], and for high-level planning and reasoning [1, 17, 44, 16, 45, 18, 6]. More recently, researchers have explored fine-tuning vision-language models (VLMs) to directly predict low-level robotic control actions, producing “vision-language-action” models (VLAs) [4, 34, 24, 23, 7, 15, 8, 53, 58, 54, 3, 2], which have demonstrated effective language following [59] and generalization to out-of-distribution test conditions and unseen semantic concepts. These works focus primarily on model development, while we focus on developing a recipe for fine-tuning such models, justifying individual design decisions with insights that we gain from our empirical analysis.

Despite the importance of fine-tuning for real-world VLA deployment, empirical analysis of effective fine-tuning recipes remains limited. While Kim et al. [23] study various parameter update strategies and from their findings show that LoRA fine-tuning enables effective adaptation to single-arm robots operating at low control frequencies (< 10 Hz), their analysis does not extend to bimanual robots with high control frequencies (25-50+ Hz), a more complex control scenario. We address this gap by exploring VLA adaptation design decisions for fast inference and reliable task execution on a real-world bimanual manipulator with a 25 Hz controller.

Recent works by Belkhale and Sadigh [2] and Pertsch et al. [38] improve VLA efficiency through new action tokenization schemes, using vector quantization or discrete cosine transform-based compression to represent action chunks (sequences of actions) with fewer tokens than simple per-dimension binning (as used in RT-2 [4] and OpenVLA [23]). While these approaches achieve 2 to $13\times$ speedups for autoregressive VLAs, we explore design decisions beyond autoregressive modeling, which remains inherently limited by iterative generation. Our parallel decoding approach, when paired with action chunking, achieves significantly greater speedups: $26\times$ to $43\times$ throughput with much lower latency (0.07 ms for single-arm tasks with one input image and 0.321 ms for bimanual tasks with three input images).

Another line of research [54, 27, 3] demonstrates effective VLA fine-tuning for high-frequency, bimanual manipulation using generative approaches like diffusion or flow match-

CITE OFT FOR / DO NOT CITE FOR

Cite for

- ✓ FT recipe: parallel + chunk + Cont + L1
- ✓ Output signature shift vs vanilla
- ✓ High-Hz FT of large VLMs w/o LL controller
- ✓ Fair ablation morals vs Cont-Diff / PD&AC
- ✓ L1 $\approx$ diffusion SR at lower cost

Do not cite for

- ✗ Universal replacement for FM heads
- ✗ BEHAVIOR q-score SOTA
- ✗ Proof L1 models multimodal demos
- ✗ Fair ALOHA win without n/rubric/JAX notes
- ✗ Drop-in household champion

BOARD TABLE · Hz / SR

Memorize these cells — Stakeholder tattoo

LIBERO		
SR avg	76.5% → 97.1%	+20.6 pp
Hz (A100)	4.2 → 109.7	26×
vs π_0 SR	π_0 94.2%	OFT wins
+wrist+proprio Hz	71.4	still >> vanilla

ALOHA (OFT+)		
Hz	1.8 → 77.9	43×
π_0 Hz still	291.6	JAX faster
Scoop	100 vs π_0 93.3	OFT+ best
Put-in-pot	51 vs π_0 42	~half; hard

Latency (tables)
LIBERO Cont-L1 ~0.073 s
ALOHA OFT+ 0.321 s
(π_0 0.086 s)

Abstract caveat
“0.07 ms / 0.321 ms”
conflicts with Table II/III
Prefer seconds

Trainable
LIBERO OFT 279M
ALOHA OFT+ 853M
(FiLM +456M)

Rule: OpenVLA/HF/LoRA ops + high Hz → OFT mandatory; stock AR obsolete for 25-50 Hz
Max wall-clock + OpenPI/JAX → $\pi_0/\pi_{0.5}$ still win latency + pretrain coverage

Visionary · close

Keep OFT in harness as open AR-family control. Deploy: $\pi_{0.5}$ /OpenPI.

Skip Discord/Drive · LIBERO 97% $\neq$ BEHAVIOR-ready

Q&A · Oct 26 · OpenVLA-OFT / 2502.19645

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